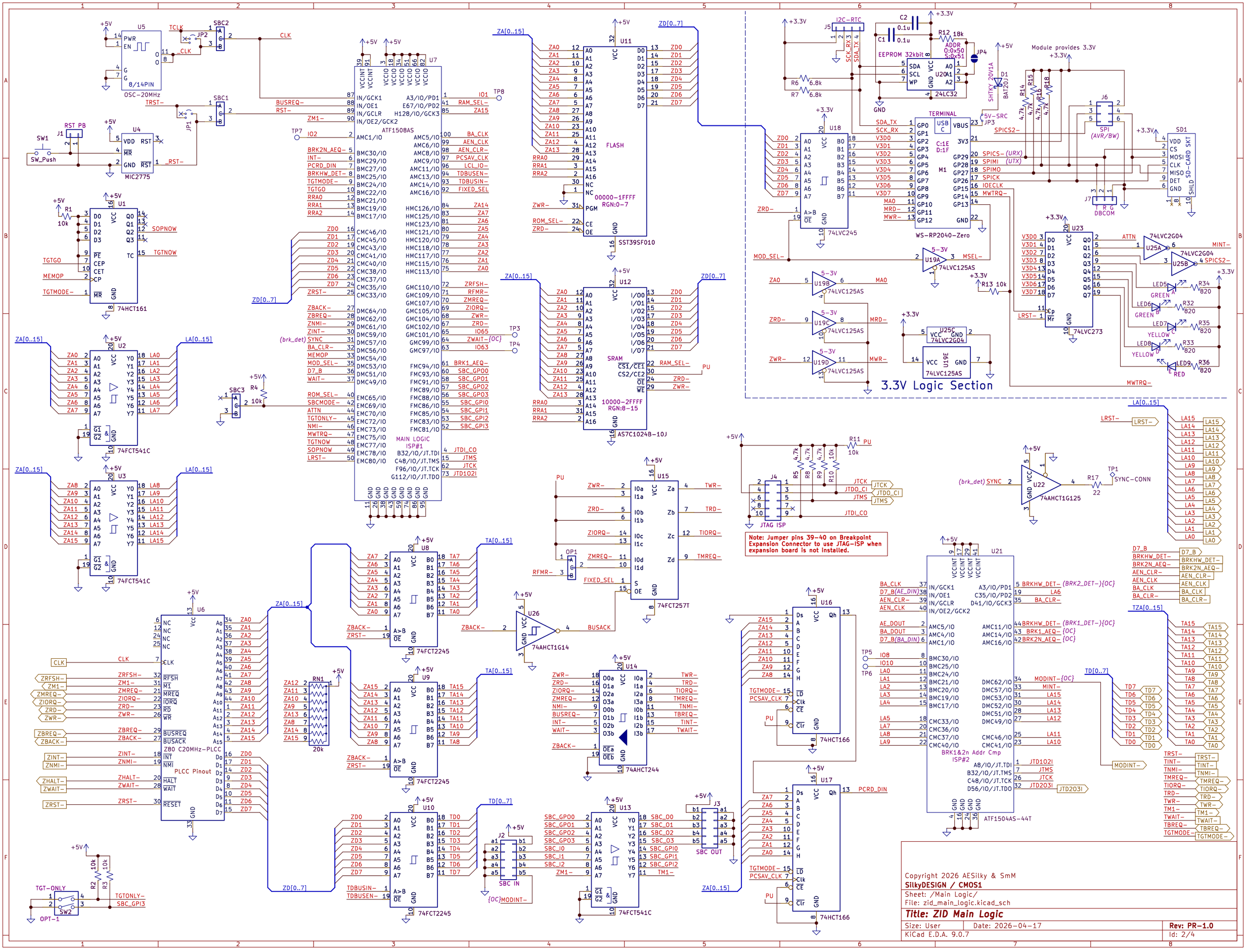
U5
PWR
EN
8/14PIN
OSC-20MHz
TCLK
JP2
SBC2
CLK
TRST-
SBC1
JP1
BUSREQ-
RST-
ZM1-
IO2
U4
VDD
RST
MR
GND
MIC2775
RST PB
J1
SW1
SW_Push
U1
74HCT161
SOPNOW
TGTNOW
TGTGO
MEMOP
TGTMODE-
U2
74FCT541C
U3
74FCT541C
LA[0..15]
ZA[0..15]
SBC3
R4
10k
ROM_SEL-
SBCMODE-
ATTN
TGTONLY-
NMI-
MWTRQ-
TGTNOW
SOPNOW
LRST-
U7
ATF1508AS
IN/GCK1
IN/OE1
IN/GCLR
IN/OE2/GCK2
AMC1/IO
BMC30/IO
BMC29/IO
BMC27/IO
BMC25/IO
BMC24/IO
BMC22/IO
BMC21/IO
BMC19/IO
BMC17/IO
CMC46/IO
CMC45/IO
CMC43/IO
CMC41/IO
CMC40/IO
CMC38/IO
CMC37/IO
CMC35/IO
CMC33/IO
DMC64/IO
DMC62/IO
DMC61/IO
DMC59/IO
DMC57/IO
DMC56/IO
DMC54/IO
DMC53/IO
DMC51/IO
DMC49/IO
EMC65/IO
EMC69/IO
EMC70/IO
EMC72/IO
EMC73/IO
EMC75/IO
EMC77/IO
EMC78/IO
EMC80/IO
A3/IO/PD1
E67/IO/PD2
H128/IO/GCK3
AMC5/IO
AMC6/IO
AMC8/IO
AMC9/IO
AMC11/IO
AMC13/IO
AMC14/IO
AMC16/IO
HMC126/IO
HMC125/IO
HMC123/IO
HMC121/IO
HMC120/IO
HMC118/IO
HMC117/IO
HMC115/IO
HMC113/IO
GMC110/IO
GMC109/IO
GMC107/IO
GMC105/IO
GMC104/IO
GMC102/IO
GMC101/IO
GMC99/IO
GMC97/IO
FMC94/IO
FMC93/IO
FMC91/IO
FMC89/IO
FMC88/IO
FMC86/IO
FMC85/IO
FMC83/IO
FMC81/IO
MAIN LOGIC
ISP#1
B32/IO/JT.TDI
C48/IO/JT.TMS
F96/IO/JT.TCK
G112/IO/JT.TDO
BA_CLK
AEN_CLK
AEN_CLR-
PCSAV_CLK
LCL_IO-
TDBUSEN-
TDBUSIN-
FIXED_SEL
ZA14
ZA7
ZA6
ZA5
ZA4
ZA3
ZA2
ZA1
ZA0
ZRFSH-
RFMR-
ZMREQ-
ZIORQ-
ZWR-
ZRD-
IO65
ZWAIT-{OC}
IO63
BRK1_AEQ-
SBC_GPO0
SBC_GPO1
SBC_GPO2
SBC_GPO3
SBC_GPI0
SBC_GPI1
SBC_GPI2
SBC_GPI3
JTDI_CO
JTMS
JTCK
JTDI02I
U11
FLASH
00000-1FFFF
RGN:0-7
SST39SF010
U12
SRAM
10000-2FFFF
RGN:8-15
AS7C1024B-10J
U15
74FCT257T
U8
74FCT2245
U9
74FCT2245
U10
74FCT2245
U26
74AHCT1G14
U14
74AHCT244
U13
74FCT541C
U6
PLCC Pinout
Z80 C20MHz-PLCC
RN1
20k
TGT-ONLY
SW2
OPT-1
R2
R3
10k
J2
SBC IN
J3
SBC OUT
{OC}MODINT-
I2C-RTC
J5
SCK_RX
SDA_TX
C1
C2
0.1u
EEPROM 32kbit
U20
24LC32
R12
18k
ADDR
0:0x50
1:0x51
JP4
R6
6.8k
R7
6.8k
GND
U18
74LVC245
TERMINAL
USB C
C:1E
D:1F
M1
WS-RP2040-Zero
SDA_TX
SCK_RX
V3D0
V3D1
V3D2
V3D3
V3D4
V3D5
V3D6
V3D7
MA0
MRD-
MWR-
MOD_SEL-
U19A
74LVC125AS
MSEL-
U19B
U19C
U19D
U19E
U25C
74LVC2G04
R13 10k
5-3V
+3.3V
3.3V Logic Section
Module provides 3.3V
D1
SHTKY 20V1A
BAI20J
5V-SRC
JP3
SPICS2-
SPICS- (URX)
SPIMI (UTX)
SPIMO
SPICK
IOECLK
MWTRQ-
J6
SPI
(AVR/BW)
J7
DBCOM
SD1
SD-CARD SKT
SHLD
R14
R15
R16
R18
4.7k
U23
74LVC273
ATTN
MINT-
SPICS2-
U25A
U25B
74LVC2G04
LED5
GREEN
LED6
GREEN
LED7
YELLOW
LED8
YELLOW
LED9
RED
R34
R32
R35
R33
R36
820
LRST-
U22
74AHCT1G125
(brk_det) SYNC
R17
22
TP1
SYNC-CONN
R11 PU
10k
R5
R8
R9
R10
JTCK
JTDO_CI
JTMS
JTDI_CO
JTAG ISP
Note: Jumper pins 39-40 on Breakpoint Expansion Connector to use JTAG-ISP when expansion board is not installed.
U21
ATF1504AS-44T
BRK1&2n Addr Cmp
ISP#2
BRKHW_DET-
BRK2N_AEQ-
BRK1_AEQ-
MODINT-
TD[0..7]
TZA[0..15]
TRST-
TINT-
TNMI-
TMREQ-
TIORQ-
TRD-
TWR-
TM1-
TWAIT-
TBREQ-
TGTMODE-
U16
74HCT166
U17
74HCT166
PCRD_DIN
PU
Copyright 2026 AESilky & SmM
SilkyDESIGN / CMOS1
Sheet: /Main Logic/
File: zid_main_logic.kicad_sch
Title: ZID Main Logic
Size: User
Date: 2026-04-17
Rev: PR-1.0
KiCad E.D.A. 9.0.7
Id: 2/4